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RRAM structure

Abstract

In some embodiments, the present disclosure relates to an integrated chip. The integrated chip includes a bottom electrode structure disposed over a lower interconnect within a lower inter-level dielectric (ILD) layer over a substrate. The bottom electrode structure has an upper surface including a noble metal. A diffusion barrier layer is over the bottom electrode structure, a data storage structure is over the diffusion barrier layer, and a top electrode structure is over the data storage structure. The diffusion barrier layer is configured to mitigate a diffusion of noble metal atoms from the bottom electrode structure to the data storage structure.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6190963	12/2000	Zhang et al.	N/A	N/A
6407422	12/2001	Asano et al.	N/A	N/A
9806255	12/2016	Hsu et al.	N/A	N/A
9954166	12/2017	Lin et al.	N/A	N/A
10043705	12/2017	Chu et al.	N/A	N/A
10193065	12/2018	Dang et al.	N/A	N/A
10910560	12/2020	Trinh	N/A	H10N 70/8833
2008/0073787	12/2007	Oh et al.	N/A	N/A
2008/0311683	12/2007	Wang	N/A	N/A
2011/0291066	12/2010	Baek et al.	N/A	N/A
2014/0061568	12/2013	Ramaswamy et al.	N/A	N/A
2014/0138347	12/2013	Toyosato et al.	N/A	N/A
2014/0192589	12/2013	Maxwell et al.	N/A	N/A
2015/0287918	12/2014	Dang et al.	N/A	N/A
2016/0064664	12/2015	Dang et al.	N/A	N/A
2016/0190305	12/2015	JangJian et al.	N/A	N/A
2016/0218283	12/2015	Trinh et al.	N/A	N/A
2017/0141300	12/2016	Trinh et al.	N/A	N/A
2017/0141305	12/2016	Yang et al.	N/A	N/A
2017/0207387	12/2016	Yang et al.	N/A	N/A
2018/0145253	12/2017	Zhou et al.	N/A	N/A
2018/0233665	12/2017	Chen et al.	N/A	N/A
2018/0269389	12/2017	Hsu et al.	N/A	N/A
2019/0305218	12/2018	Trinh et al.	N/A	N/A

OTHER PUBLICATIONS

Wong, et al. "Metal-Oxide RRAM." Proceedings of the IEEE, vol. 100, No. 6, Jun. 2012. cited by applicant

U.S. Appl. No. 15/939,864, filed Mar. 29, 2018. cited by applicant

Non-Final Office Action dated Dec. 30, 2019 for U.S. Appl. No. 16/232,342. cited by applicant

Final Office Action dated Jun. 23, 2020 for U.S. Appl. No. 16/232,342. cited by applicant

Notice of Allowance dated Oct. 5, 2020 for U.S. Appl. No. 16/232,342. cited by applicant

Non-Final Office Action dated Jan. 21, 2022 for U.S. Appl. No. 17/142,591. cited by applicant

Notice of Allowance dated Jun. 10, 2022 for U.S. Appl. No. 17/142,591. cited by applicant

Non-Final Office Action dated Aug. 23, 2023 for U.S. Appl. No. 17/874,406. cited by applicant

Notice of Allowance dated Dec. 20, 2023 for U.S. Appl. No. 17/874,406. cited by applicant

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Background/Summary

REFERENCE TO RELATED APPLICATIONS (1) This Application is a Divisional of U.S. application Ser. No. 17/874,406, filed on Jul. 27, 2022, which is a Continuation of U.S. application Ser. No. 17/142,591, filed on Jan. 6, 2021 (now U.S. Pat. No. 11,482,668, issued on Oct. 25, 2022), which is a Divisional of U.S. application Ser. No. 16/232,342, filed on Dec. 26, 2018 (now U.S. Pat. No. 10,910,560, issued on Feb. 2, 2021), which claims the benefit of U.S. Provisional Application No. 62/734,575, filed on Sep. 21, 2018. The contents of the above-referenced Patent Applications are hereby incorporated by reference in their entirety.

BACKGROUND

(1) Many modern-day electronic devices contain electronic memory configured to store data. Electronic memory may be volatile memory or non-volatile memory. Volatile memory stores data when it is powered, while non-volatile memory is able to store data when power is removed. Resistive random-access memory (RRAM) devices are one promising candidate for a next generation non-volatile memory technology. This is because RRAM devices provide for many advantages, including a fast write time, high endurance, low power consumption, and low susceptibility to damage from radiation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a cross-sectional view of some embodiments of an integrated chip having a disclosed resistive random-access memory (RRAM) device comprising a diffusion barrier layer disposed between a bottom electrode and a data storage layer.

(3) FIG. 2 illustrates a cross-sectional view of some additional embodiments of an integrated chip having a disclosed RRAM device comprising a diffusion barrier layer.

(4) FIGS. 3A-3B illustrate some additional embodiments of an integrated chip having a disclosed RRAM device comprising a diffusion barrier layer.

(5) FIGS. 4A-4B illustrate some additional embodiments of an integrated chip having an embedded RRAM device comprising a diffusion barrier layer disposed between a bottom electrode and a data storage layer.

(6) FIGS. 5-6 illustrate cross-sectional views of some additional embodiments of an integrated chip having a disclosed RRAM device comprising a diffusion barrier layer.

(7) FIGS. 7-17 illustrate cross-sectional views of some embodiments of a method of forming an integrated chip having a disclosed RRAM device comprising a diffusion barrier layer disposed between a bottom electrode and a data storage layer.

(8) FIG. 18 illustrates a flow diagram of some embodiments of a method of forming an integrated chip having a disclosed RRAM device comprising a diffusion barrier layer disposed between a bottom electrode and a data storage layer.

DETAILED DESCRIPTION

(9) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(10) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(11) Resistive random access memory (RRAM) devices generally comprise a data storage layer (e.g., a layer of high-k dielectric material) arranged between conductive bottom and top electrodes disposed within a back-end-of-the-line (BEOL) metallization stack. RRAM devices are configured to operate based upon a process of reversible switching between resistive states. This reversible switching is enabled by selectively forming a conductive filament through the data storage layer. For example, the data storage layer, which is normally insulating, can be made to conduct by applying a voltage across the conductive electrodes to form a conductive filament extending through the data storage layer. An RRAM device having a first (e.g., high) resistive state corresponds to a first data value (e.g., a logical ‘0’) and an RRAM device having a second (e.g., low) resistive state corresponds to a second data value (e.g., a logical ‘1’).

(12) To improve a reliability of an RRAM device, the bottom electrode of an RRAM device may comprise a noble metal (i.e., an inert metal). Because a noble metal has a low reactivity, the use of a noble metal in the bottom electrode of the RRAM device may prevent degradation of the bottom electrode due to interactions with ions (e.g., oxygen ions) from an overlying data storage layer. However, it has been appreciated that metal atoms from a noble metal can diffuse into an overlying data storage layer during high temperature processes used during the fabrication of overlying layers of an RRAM device. The diffusion of noble metal atoms from the bottom electrode forms hillocks (i.e., protrusions) along a top of the bottom electrode. The hillocks comprise a combination of the noble metal atoms and the data storage layer (e.g., hafnium titanium oxide). The hillocks extend into the data storage layer, thereby effectively thinning the data storage layer and leading to a

reduced reliability of the resulting RRAM device and/or leakage between the bottom and top electrodes of the resulting RRAM device.

(13) The present disclosure, in some embodiments, relates to an RRAM device comprising a bottom electrode separated from a top electrode by a data storage layer. A diffusion barrier layer is arranged between the bottom electrode and the data storage layer. The diffusion barrier layer is configured to prevent the diffusion of metal atoms (e.g., noble metal atoms) from the bottom electrode into the data storage layer and to thereby prevent the formation of hillocks within the data storage layer. Preventing the diffusion of metal atoms from the bottom electrode into the data storage layer, allows for the data storage layer to have a substantially uniform thickness that increases a reliability of the RRAM device.

(14) FIG. 1 illustrates a cross-sectional view of some embodiments of an integrated chip **100** comprising a resistive random-access memory (RRAM) device having a diffusion barrier layer disposed between a bottom electrode and a data storage layer.

(15) The integrated chip **100** comprises an RRAM device **111** disposed within a dielectric structure **106** over a substrate **102**. The RRAM device **111** is separated from the substrate **102** by one or more lower interconnect layers **108** disposed within the dielectric structure **106**. In some embodiments, the one or more lower interconnect layers **108** may comprise conductive contacts **107** coupled to overlying and alternating layers of interconnect vias **109** and interconnect wires **110**. The one or more lower interconnect layers **108** are configured to couple the RRAM device **111** to an access transistor **104** arranged within the substrate **102**.

(16) The RRAM device **111** comprises a dielectric data storage layer **116** arranged between a bottom electrode **112** and a top electrode **118**. The bottom electrode **112** is coupled to the one or more lower interconnect layers **108** and the top electrode **118** is coupled to an upper interconnect structure **120** comprising an interconnect wire or via. The dielectric data storage layer **116** is configured to store data states by undergoing reversible changes between a high resistive state associated with a first data state (e.g., a '0') and a low resistive state associated with a second data state (e.g., a '1'). For example, during operation, to achieve a low resistive state within the dielectric data storage layer **116**, a first set of bias conditions may be applied to the bottom electrode **112** and the top electrode **118**. The first set of bias conditions drive oxygen from dielectric data storage layer **116** to the top electrode **118**, thereby forming a conductive filament **117** of oxygen vacancies across the dielectric data storage layer **116**. Alternatively, to achieve a high resistive state within the dielectric data storage layer **116**, a second set of bias conditions may be applied to the bottom electrode **112** and the top electrode **118**. The second set of bias conditions break the conductive filament **117** by driving oxygen from the top electrode **118** to the dielectric data storage layer **116**.

(17) A diffusion barrier layer **114** is arranged between the bottom electrode **112** and the dielectric data storage layer **116**. The diffusion barrier layer **114** is configured to prevent the diffusion of metal atoms (e.g., noble metal atoms) from the bottom electrode **112** towards the dielectric data storage layer **116**. By preventing the diffusion of metal atoms from the bottom electrode **112** towards the dielectric data storage layer **116**, hillocks comprising the metal atoms are prevented from forming along a top of the bottom electrode **112**. By preventing the formation of hillocks along the top of the bottom electrode **112**, a uniformity of a thickness of the dielectric data storage layer **116** is improved, thereby improving performance and/or reliability of the RRAM device **111**. The diffusion barrier layer **114** also improves a variation in thicknesses of dielectric data storage layers between RRAM cells, thereby improving cell-to-cell variation and die yield.

(18) FIG. 2 illustrates an additional embodiment of a cross-sectional view of an integrated chip **200** having an RRAM device.

(19) The integrated chip **200** comprises an RRAM device **111** disposed within a dielectric structure **106** arranged over a substrate **102**. In some embodiments, the dielectric structure **106** comprises a plurality of stacked inter-level dielectric (ILD) layers **202a-202d**. The plurality of stacked ILD

layers **202a-202d** laterally surround one or more lower interconnect layers **108** comprising interconnect vias **109** and interconnect wires **110**. The one or more lower interconnect layers **108** are configured to couple the RRAM device **111** to an access transistor **104** arranged within the substrate **102**. In some embodiments, the access transistor **104** may comprise a MOSFET device having a gate electrode **104d** that is arranged between a source region **104a** and a drain region **104b** and that is separated from the substrate **102** by a gate dielectric **104c**.

(20) The RRAM device **111** comprises a dielectric data storage layer **116** arranged between a bottom electrode **112** and a top electrode **118**. The bottom electrode **112** is arranged over the one or more lower interconnect layers **108** and is laterally surrounded by a lower insulating layer **208**. In various embodiments, the lower insulating layer **208** may comprise silicon nitride, silicon dioxide, silicon carbide, or the like. In some embodiments, the dielectric data storage layer **116** may have a bottom surface having a first width and a top surface having a second width that is less than the first width. In such embodiments, the dielectric data storage layer **116** may have a lower sidewall coupled to an upper sidewall by a horizontally extending surface that overlies the dielectric data storage layer **116**. In some embodiments, the dielectric data storage layer **116** may have a thickness in a range of between approximately 30 Angstroms and approximately 100 Angstroms. In other embodiments, the dielectric data storage layer **116** may have a thickness in a range of between approximately 25 Angstroms and approximately 75 Angstroms. In some embodiments, the top electrode **118** may have a thickness in a range of between approximately 50 Angstroms and approximately 200 Angstroms.

(21) In some embodiments, the bottom electrode **112** may have a bottom electrode diffusion barrier **112a** and a bottom electrode metal **112b** arranged over the bottom electrode diffusion barrier **112a**. The bottom electrode diffusion barrier **112a** is laterally surrounded by the lower insulating layer **208**. The bottom electrode diffusion barrier **112a** fills an opening defined by sidewalls of the lower insulating layer **208**. In some embodiments, the bottom electrode diffusion barrier **112a** may extend from within the opening to over the lower insulating layer **208**. In such embodiments, the bottom electrode diffusion barrier **112a** may have a first thickness between the sidewalls of the lower insulating layer **208** and a smaller second thickness over the lower insulating layer **208**. In some embodiments, the first thickness may be in a range of between approximately 200 Angstroms and approximately 400 Angstroms. In some embodiments, the bottom electrode metal **112b** may have a thickness in a range of between approximately 50 Angstroms and approximately 300 Angstroms. In some embodiments, the bottom electrode diffusion barrier **112a** barrier has a curved lower sidewall surrounded by the lower insulating layer **208** and a substantially flat upper sidewall over the lower insulating layer **208**. In some embodiments, the bottom electrode diffusion barrier **112a** may have a substantially flat upper surface.

(22) In some embodiments, the bottom electrode metal **112b** may comprise a first conductive material and the bottom electrode diffusion barrier **112a** and the top electrode **118** may comprise one or more conductive materials that are different than the first conductive material. For example, in some embodiments, the bottom electrode metal **112b** may comprise a noble metal, such as platinum, ruthenium, iridium, gold, or the like. The noble metal provides the RRAM device **111** with good endurance and data retention. This is because the noble metal has a low reactivity with ions (e.g., oxygen ions), and thus can help to prevent the permeability of oxygen ions into the bottom electrode **112** during RRAM cycling. The bottom electrode diffusion barrier **112a** and the top electrode **118** may comprise a metal, such as titanium, titanium nitride, tantalum, tantalum nitride, tungsten, tungsten nitride, or the like. In some embodiments, the bottom electrode diffusion barrier **112a** and the top electrode **118** may comprise a same material.

(23) A diffusion barrier layer **114** is arranged between the bottom electrode **112** and a dielectric data storage layer **116**. The diffusion barrier layer **114** is vertically separated from the bottom electrode diffusion barrier **112a** by way of the bottom electrode metal **112b**. In some embodiments, the diffusion barrier layer **114** contacts the bottom electrode metal **112b** along a substantially flat

interface. The diffusion barrier layer **114** may comprise a different material than the bottom electrode metal **112b**. In some embodiments, the diffusion barrier layer **114** may comprise tantalum nitride, titanium nitride, titanium carbon nitride, tungsten nitride, or the like. In other embodiments, the diffusion barrier layer **114** may comprise a noble metal oxide (e.g., ruthenium oxide, iridium oxide, rhodium oxide, or the like) or a noble metal nitride (e.g., ruthenium nitride, iridium nitride, or the like).

(24) The diffusion barrier layer **114** has a thickness $t_{sub.1}$ that is sufficient to prevent the formation of hillocks comprising metal atoms (e.g., noble metal atoms) of the bottom electrode **112** from extending into the dielectric data storage layer **116**. In some embodiments, the thickness $t_{sub.1}$ is in a range of between approximately 5 Angstroms and approximately 30 Angstroms. A thickness of the diffusion barrier layer **114** that is less than 5 Angstroms may be unable to effectively prevent diffusion of metal atoms from the bottom electrode metal **112b** in a reliable manner, while a thickness of the diffusion barrier layer **114** that is greater than 30 Angstroms may interfere with performance of the RRAM device **111** by mitigating the advantages using a noble metal within the bottom electrode **112**. In other embodiments, the thickness $t_{sub.1}$ may have different thicknesses.

(25) A capping layer **210** is arranged between the dielectric data storage layer **116** and the top electrode **118**. The capping layer **210** is configured to store oxygen, which can facilitate resistive changes within the dielectric data storage layer **116**. In various embodiments, the capping layer **210** may comprise hafnium, titanium, tantalum, aluminum, zirconium, or the like. In some embodiments, the capping layer **210** may have a thickness in a range of between approximately 70 Angstroms and approximately 200 Angstroms. In some embodiments, a dielectric spacer **212** may be arranged along sidewalls of the capping layer **210** and the top electrode **118**. In some embodiments, the dielectric spacer **212** may also be arranged along a sidewall of the dielectric data storage layer **116**. In some embodiments, the dielectric spacer **212** may be arranged on a horizontally extending surface of the top electrode **118** and/or the dielectric data storage layer **116**. In some embodiments, the dielectric spacer **212** may comprise a nitride (e.g., silicon nitride), an oxide (e.g., silicon oxide), a carbide (e.g., silicon carbide), or the like.

(26) In some embodiments, sidewalls of the diffusion barrier layer **114** may be angled at a first angle α with respect to an upper surface of the lower insulating layer **208**. The first angle α may be greater than 90° . In some embodiments, the sidewalls of the diffusion barrier layer **114** may be aligned along a line with sidewalls of the bottom electrode **112** and the dielectric data storage layer **116**. In some embodiments, the sidewalls of the capping layer **210** may be angled at a second angle β with respect to an upper surface of the dielectric data storage layer **116**. The second angle β may also be greater than 90° . In some embodiments, the first angle α is different (e.g., greater than) the second angle β .

(27) FIG. 3A illustrates a cross-sectional view of an additional embodiment of an integrated chip **300** having an RRAM device.

(28) The integrated chip **300** comprises an RRAM device **301** arranged over a substrate **102**. The RRAM device **301** comprises a dielectric data storage layer **116** arranged between a bottom electrode **112** and a top electrode **118**. The bottom electrode **112** comprises a bottom electrode diffusion barrier **112a** and a bottom electrode metal **112b** over the bottom electrode diffusion barrier **112a**. A capping layer **210** may be arranged between the dielectric data storage layer **116** and the top electrode **118**.

(29) The bottom electrode diffusion barrier **112a**, the bottom electrode metal **112b**, the dielectric data storage layer **116**, the capping layer **210**, and the top electrode **118** respectively have an inner region **302** laterally surrounded by an outer region **304**. The inner region **302** has a recessed upper surface arranged laterally between and vertically below upper surfaces of the outer region **304**. As shown in top-view **306** of FIG. 3B, the outer region **304** of the bottom electrode diffusion barrier **112a** extends along an outermost perimeter of the bottom electrode diffusion barrier **112a**. In some

embodiments, the outer region **304** may continually extend in an unbroken ring around the inner region **302**.

(30) In some embodiments, the bottom electrode diffusion barrier **112a**, the bottom electrode metal **112b**, the diffusion barrier layer **114**, the capping layer **210**, and the top electrode **118** may respectively have a substantially equal thickness between outermost sidewalls. For example, the bottom electrode diffusion barrier **112a** may have a substantially equal thickness in a range of between approximately 30 Angstroms and approximately 200 Angstroms. In some embodiments, the inner region **302** of the dielectric data storage layer **116** may have a first thickness and the outer region of the dielectric data storage layer **116** may have a second thickness that is less than the first thickness.

(31) FIG. 4A illustrates a cross-sectional view of some additional embodiments of an integrated chip **400** having an RRAM device.

(32) The integrated chip **400** comprises a substrate **102** including a logic region **402** and an embedded memory region **404**. As shown in top-view **416** of FIG. 4B, in some embodiments, the logic region **402** may surround the embedded memory region **404**.

(33) Referring to FIG. 4A, a dielectric structure **106** is arranged over the substrate **102**. The dielectric structure **106** comprises a plurality of stacked ILD layers **202a-202d** separated by etch stop layers **406**. In some embodiments, the plurality of stacked ILD layers **202a-202d** may comprise one or more of an oxide layer, a low-k dielectric layer, an ultra low-k dielectric layer, or the like. In some embodiments, the etch stop layers **406** may comprise a nitride (e.g., silicon nitride), a carbide (e.g., silicon carbide), or the like.

(34) The logic region **402** comprises a transistor device **408** arranged within the substrate **102**. The transistor device **408** comprises a source region **408a**, a drain region **408b** separated from the source region **408a** by a channel region, and a gate structure **408g** over the channel region. In some embodiments, the transistor device **408** may comprise a high-k metal gate (HKMG) transistor. In such embodiments, the gate structure **408g** may comprise a metal gate electrode (e.g., comprising aluminum, ruthenium, palladium, or the like) and a gate dielectric comprising a high-k dielectric (e.g., comprising aluminum oxide, hafnium oxide, or the like). In other embodiments, the gate structure **408g** may comprise a polysilicon gate electrode and a gate dielectric comprising an oxide (e.g., silicon dioxide).

(35) The source region **408a** is coupled to a plurality of interconnect layers surrounded by the dielectric structure **106**. The plurality of interconnect layers comprise conductive contacts **410**, interconnect wires **412**, and interconnect vias **414**. In some embodiments, the plurality of interconnect layers may comprise copper, tungsten, aluminum, and/or the like.

(36) The embedded memory region **404** comprises an access transistor **104** arranged within the substrate **102**. The access transistor **104** is coupled to an RRAM device **111**. The RRAM device **111** is arranged along a horizontal plane that intersects one of the interconnect vias **414** within the logic region **402**. In some embodiments, one or more isolation structures **405** may be arranged within the substrate **102** on opposing sides of the access transistor **104**. The isolation structures **405** may comprise one or more dielectric materials arranged within trenches defined by interior surfaces of the substrate **102**. In some embodiments, the isolation structures **405** may comprise shallow trench isolation (STI) structures. In some such embodiments, the isolation structures **405** may comprise a same isolation structure continuously extending in a closed loop around a perimeter of the access transistor **104**.

(37) FIG. 5 illustrates some additional embodiments of an integrated chip **500** having an RRAM device.

(38) The integrated chip **500** comprises a 1T1R RRAM cell architecture having an access transistor **104** connected to an RRAM device **111**. The access transistor **104** is arranged within a substrate **102**. In some embodiments, the access transistor **104** may comprise a MOSFET device having a gate electrode **104d** that is arranged between a source region **104a** and a drain region **104b** and that

is separated from the substrate by a gate dielectric **104c**.

(39) A dielectric structure **106** is arranged over the substrate **102**. One or more lower interconnect layers **108** including conductive contacts **107**, interconnect vias **109**, and interconnect wires **110**, are surrounded by the dielectric structure **106**. The interconnect wires **110** include a source-line SL comprising a first interconnect wire that is electrically coupled to the source region **104a**. In some embodiments, the source-line SL may be arranged in a second interconnect wire layer that is connected to the source region **104a** through a conductive contact, a first interconnect wire, and a first interconnect via. The interconnect wires **110** further comprise a word-line WL comprising a second interconnect wire that is electrically coupled to the gate electrode **104d**. In some embodiments, the word-line WL may be arranged in the first interconnect wire layer that is connected to the gate electrode **104d** by way of a conductive contact.

(40) An RRAM device **111** is arranged over the dielectric structure **106**. The RRAM device comprises a bottom electrode **112** separated from a top electrode **118** by a diffusion barrier layer **114**, a dielectric data storage layer **116**, and a capping layer **210**. The bottom electrode **112** is directly connected to the drain region **104b** by the one or more lower interconnect layers **108**. The top electrode **118** is further coupled to a bit-line BL by way of an upper interconnect structure **120**.

(41) In some embodiments, dielectric spacers **212** are arranged along opposing sides of the top electrode **118**. In some embodiments, the dielectric spacers **212** may have a horizontally extending segment **212a** that protrudes outward from a sidewall of the dielectric spacers **212**. In various embodiments, horizontally extending segments may protrude outward from opposing sides of the dielectric spacers **212** or a horizontally extending segment may protrude outward from one side of the dielectric spacers **212** and not from an opposing side of the dielectric spacers **212**.

(42) Although integrated chip **500** illustrates the word-line WL, the source-line SL, the bit-line BL, and the RRAM device **111** as being located at certain levels within a BEOL (back-end-of-the-line) stack, it will be appreciated that the position of these elements is not limited to those illustrated positions. Rather, the elements may be at different locations within a BEOL stack. For example, in some alternative embodiments, the RRAM device **111** may be located between a second and third metal interconnect wire.

(43) FIG. **6** illustrates some additional embodiments of an integrated chip **600** having an RRAM device.

(44) The integrated chip **600** comprises a 2T1R RRAM cell architecture having a first access transistor **602** and a second access transistor **604** connected in parallel to an RRAM device **111**. The first access transistor **602** and the second access transistor **604** are arranged within a substrate **102**. In some embodiments, the first access transistor **602** has a first gate electrode **602a** that is arranged between a first source region **602b** and a common drain region **603**. The first source region **602b** is coupled to a source-line SL and the first gate electrode **602a** is coupled to a word-line WL. The second access transistor **604** has a second gate electrode **604a** that is arranged between a second source region **604b** and the common drain region **603**. The second source region **604b** is coupled to the source-line SL and the second gate electrode **604a** is coupled to the word-line WL.

(45) One or more lower interconnect layers **108** couple the common drain region **603** to the RRAM device **111** arranged within a dielectric structure **106** over the substrate **102**. By coupling the common drain region **603** to the RRAM device **111**, a driving current provided to the RRAM device **111** can be increased over RRAM cells that utilize a single access transistor (e.g., as shown in FIG. **5**).

(46) FIGS. **7-17** illustrate cross-sectional views **700-1700** of some embodiments of a method of forming an integrated chip having an RRAM cell comprising a diffusion barrier layer disposed between a bottom electrode and a data storage layer. Although FIGS. **7-17** are described in relation to a method, it will be appreciated that the structures disclosed in FIGS. **7-17** are not limited to such a method, but instead may stand alone as structures independent of the method.

(47) As shown in cross-sectional view **700** of FIG. 7, a substrate **102** is provided. In various embodiments, the substrate **102** may comprise any type of semiconductor body (e.g., silicon/CMOS bulk, SiGe, SOI, etc.) such as a semiconductor wafer or one or more die on a wafer, as well as any other type of semiconductor and/or epitaxial layers formed thereon and/or otherwise associated therewith. In some embodiments, one or more isolation structures **405** (e.g., shallow trench isolation structures) may be formed within the substrate **102**. The one or more isolation structures **405** may be formed by selectively etching the substrate **102** to form a trench **702** defined by sidewalls of the substrate **102**. The trench **702** is subsequently filled with one or more dielectric materials.

(48) An access transistor **104** is formed over the substrate **102**. In some embodiments, the access transistor **104** may be formed by forming a gate dielectric over the substrate and forming a gate material over the gate dielectric. The gate dielectric and the gate material may be formed by way of vapor deposition processes (e.g., CVD, PE-CVD, PVD, or ALD). In some embodiments, the gate material may comprise doped polysilicon. In some embodiments, the gate material may comprise a sacrificial gate material that is subsequently replaced with a metal gate material, such as aluminum, cobalt, ruthenium, or the like.

(49) The gate dielectric and the gate material are patterned to define a gate structure having a gate dielectric **104c** and a gate electrode **104d** over the gate dielectric **104c**. In some embodiments, the gate dielectric and the gate material may be selectively patterned according to a masking layer (not shown) formed over the gate material. In some embodiments, the masking layer may comprise a photosensitive material (e.g., photoresist) formed by a spin coating process. In such embodiments, the photosensitive material is selectively exposed to electromagnetic radiation according to a photomask. The electromagnetic radiation modifies a solubility of exposed regions within the photosensitive material to define soluble regions. The photosensitive material is subsequently developed to define openings within the photosensitive material by removing the soluble regions. In other embodiments, the masking layer may comprise a hard mask layer (e.g., a silicon nitride layer, a silicon carbide layer, or the like).

(50) As shown in cross-sectional view **800** of FIG. 8, one or more lower interconnect layers **108** are formed within one or more stacked lower inter-level dielectric (ILD) layers **202a-202c** (e.g., an oxide, a low-k dielectric, or an ultra low-k dielectric) over the substrate **102**. In some embodiments, the one or more lower interconnect layers **108** may be respectively formed by selectively etching one of the one or more stacked lower ILD layers **202a-202c** to define an opening within the ILD layer. A conductive material (e.g., copper, aluminum, etc.) is then deposited to fill the opening, and a planarization process (e.g., a chemical mechanical planarization process) is performed to remove excess conductive material from over the ILD layer.

(51) As shown in cross-sectional view **900** of FIG. 9, a lower insulating layer **208** is formed onto the one or more lower interconnect layers **108** and the one or more stacked lower ILD layers **202a-202c**. In some embodiments, the lower insulating layer **208** may comprise silicon-nitride (SiN), silicon-carbide (SiC), or a similar composite dielectric film. In some embodiments, the lower insulating layer **208** may be formed by a deposition technique (e.g., physical vapor deposition (PVD), chemical vapor deposition (CVD), PE-CVD, atomic layer deposition (ALD), sputtering, etc.) to a thickness in a range of between approximately 50 Angstroms and approximately 250 Angstroms. After being deposited, the lower insulating layer **208** is selectively exposed to a first etchant **902** (e.g., a dry etchant, a wet etchant) that forms sidewalls defining an opening **904** within the lower insulating layer **208**. The opening **904** extends through the lower insulating layer **208** to one of the one or more lower interconnect layers **108**. In some embodiments, the lower insulating layer **208** may be selectively exposed to the first etchant **902** according to a masking layer (not shown) formed over the lower insulating layer **208**.

(52) As shown in cross-sectional view **1000** of FIG. 10, a bottom electrode structure **1002** is formed over the one or more lower interconnect layers **108** and the lower insulating layer **208**. The

bottom electrode structure **1002** extends from within the opening **904** to a position overlying the lower insulating layer **208**. In some embodiments, the bottom electrode structure **1002** is formed by performing separate depositions to form a bottom electrode barrier film **1002a** and to subsequently form a bottom electrode metal film **1002b** over the bottom electrode barrier film **1002a**. In some embodiments, the separate depositions may be performed in-situ (e.g., without breaking a vacuum of a processing chamber in which the depositions are performed). In some embodiments, the bottom electrode barrier film **1002a** may comprise tantalum nitride (TaN), titanium nitride (TiN), tungsten (W), tungsten nitride (WN), or the like. In some embodiments, the bottom electrode metal film **1002b** may comprise a noble metal, such as platinum (Pt), ruthenium (Ru), iridium (Ir), gold (Au), or the like.

(53) As shown in cross-sectional view **1100** of FIG. **11**, a diffusion barrier film **1102** is formed on the bottom electrode structure **1002**. The diffusion barrier film **1102** is formed to a thickness in a range of between approximately 5 Angstroms and approximately 30 Angstroms. In some embodiments, the diffusion barrier film **1102** may comprise tantalum nitride (TaN), titanium nitride (TiN), titanium carbon nitride (TiCN), tungsten nitride (WN), or the like. In some additional embodiments, the diffusion barrier film **1102** may comprise a noble metal oxide (e.g., ruthenium oxide, iridium oxide, platinum oxide, palladium oxide, or the like) or a noble metal nitride (e.g., ruthenium nitride, iridium nitride, or the like).

(54) In some embodiments, the diffusion barrier film **1102** maybe formed by way of a deposition technique (e.g., PVD, CVD, PE-CVD, ALD, sputtering, etc.). In other embodiments, the diffusion barrier film **1102** maybe formed by exposing a noble metal on the bottom electrode **112** to the oxygen gas (e.g., O.sub.3) at an elevated temperature (e.g., a temperature of greater than or equal to approximately 150° C.) to form a noble metal oxide. In yet other embodiments, the diffusion barrier film **1102** maybe formed by exposing a noble metal on the bottom electrode **112** to a nitrogen based plasma (e.g., N.sub.2O plasma) to form a noble metal nitride.

(55) In some embodiments, the diffusion barrier film **1102** maybe formed in-situ (i.e., without breaking a vacuum) with the bottom electrode metal film **1002b**. In such embodiments, the diffusion barrier film **1102** is formed in direct contact with the bottom electrode metal film **1002b**. In other embodiments, the diffusion barrier film **1102** maybe formed ex-situ with the bottom electrode metal film **1002b**. In such embodiments, an oxide film (e.g., having a thickness of approximately 5 Angstroms or less) may form onto a top of the bottom electrode metal film **1002b**. In some such embodiments, the diffusion barrier film **1102** maybe separated from the bottom electrode metal film **1002b** by the oxide film. In other such embodiments, a hydrofluoric acid dip may be performed prior to the deposition of the diffusion barrier film **1102** to remove the oxide film.

(56) As shown in cross-sectional view **1200** of FIG. **12**, a dielectric data storage film **1202** is formed over the diffusion barrier film **1102**, a capping film **1204** is formed over the dielectric data storage film **1202**, and a top electrode structure **1206** is formed over the capping film **1204**. In some embodiments, the dielectric data storage film **1202** may comprise a high-k dielectric material having a variable resistance. For example, in some embodiments, the dielectric data storage film **1202** may comprise hafnium oxide (HfO.sub.X), zirconium oxide (ZrO.sub.X), aluminum oxide (AlO.sub.X), nickel oxide (NiO.sub.X), tantalum oxide (TaO.sub.X), titanium oxide (TiO.sub.X), or the like. In some embodiments, the capping film **1204** may comprise a metal (e.g., such as titanium (Ti), hafnium (Hf), platinum (Pt), aluminum (Al), or the like) or a metal oxide (e.g., such as titanium oxide (TiO.sub.x), hafnium oxide (HfO.sub.x), zirconium oxide (ZrO.sub.x), germanium oxide (GeO.sub.x), cesium oxide (CeO.sub.x), or the like). In some embodiments, the top electrode structure **1206** may comprise a metal, such as titanium (Ti), tantalum (Ta), or the like.

(57) As shown in cross-sectional view **1300** of FIG. **13**, a first RRAM patterning process is performed. The first RRAM patterning process selectively patterns the top electrode structure (**1206** of FIG. **12**) to define a top electrode **118**. The first RRAM patterning process may also

selectively pattern the capping film (**1204** of FIG. **12**) to define a capping layer **210**. In some embodiments, the first patterning process comprises forming a hard mask layer **1304** over the top electrode structure (**1206** of FIG. **12**). The top electrode structure is then exposed to a second etchant **1302** (e.g., a dry etchant) according to the hard mask layer **1304** to remove unmasked parts of the capping film (**1204** of FIG. **12**) and the top electrode structure (**1206** of FIG. **12**). In various embodiments, the hard mask layer **1304** may comprise silicon-oxide (SiO_2), silicon-oxynitride (SiON), silicon-nitride (SiN), silicon-carbide (SiC), or the like.

(58) In some embodiments, the first RRAM patterning process may reduce a thickness of outer regions of the dielectric data storage film **1202**, so that the dielectric data storage film **1202** has a central region with a first thickness and outer regions with a second thickness that is less than the first thickness. In some additional embodiments (not shown), the first patterned process may etch through the dielectric data storage film **1202** and into the diffusion barrier film **1102**. In such embodiments, the diffusion barrier film **1102** may have a central region with a thickness that is greater than that of outer regions. In some such embodiments, due to different etch selectivities of the diffusion barrier film **1102** and the dielectric data storage film **1202**, outermost sidewalls of the dielectric data storage film **1202** may be laterally offset by a non-zero distance from an uppermost sidewall defining the central region of the diffusion barrier film **1102**. In some embodiments, the hard mask layer **1304** may be removed after the first RRAM patterning process is completed.

(59) As shown in cross-sectional view **1400** of FIG. **14**, in some embodiments, dielectric spacers **212** may be formed on opposing sides of the top electrode **118** and the capping layer **210**. The dielectric spacers **212** may also extend over an upper surface of the top electrode **118**. In various embodiments, the dielectric spacers **212** may comprise silicon nitride, silicon dioxide (SiO_2), silicon oxy-nitride (e.g., SiON), or the like. In some embodiments, the dielectric spacers **212** may have outermost sidewalls that are oriented at an angle Ω with respect to a sidewall of the diffusion barrier layer **114**. In some embodiments, the angle Ω may be in a range that is between 90° and 180° .

(60) The dielectric spacers **212** may be formed by depositing a spacer layer on the dielectric data storage film (**1202** of FIG. **13**), the capping layer **210**, the top electrode **118**, and the hard mask layer **1304**. In some embodiments, the spacer layer may be deposited by a deposition technique (e.g., PVD, CVD, PE-CVD, ALD, sputtering, etc.) to a thickness in a range of between approximately 100 angstroms and approximately 500 angstroms. The spacer layer is subsequently patterned (e.g., using a lithography process) to define the dielectric spacers **212** along opposing sides of the top electrode **118** and over the top electrode **118**. In some embodiments, wherein the second etchant (**1302** of FIG. **13**) etches through the dielectric data storage layer **116**, the dielectric spacers **212** may contact sidewalls of diffusion barrier layer **114**.

(61) In some embodiments, the separate patterning process used to define the dielectric spacers **212** may result in the dielectric spacers **212** having a horizontally extending segment **212a** that protrudes outward from a sidewall of the dielectric spacers **212**. In some embodiments, horizontally extending segments may protrude outward from opposing sides of the dielectric spacers **212**, while in other embodiments (not shown) the horizontally extending segment may protrude outward from one side of the dielectric spacers **212** and not from an opposing side of the dielectric spacers **212**.

(62) After formation of the dielectric spacers **212**, a second RRAM patterning process is performed. The second RRAM patterning process defines an RRAM device **111** by selectively removing the dielectric data storage film (**1202** of FIG. **13**), the diffusion barrier film (**1102** of FIG. **13**), and the bottom electrode structure (**1002** of FIG. **13**) to respectively define a dielectric data storage layer **116**, a diffusion barrier layer **114**, and a bottom electrode **112**. In some embodiments, the second RRAM patterning process selectively exposes the dielectric data storage film (**1202** of FIG. **13**), the diffusion barrier film (**1102** of FIG. **13**), and the bottom electrode structure (**1002** of FIG. **13**) to a third etchant **1402** according to a mask comprising the dielectric spacers **212**.

(63) As shown in cross-sectional view **1500** of FIG. **15**, an upper insulating layer **1502** may be

formed over the RRAM device **111**. In various embodiments, the upper insulating layer **1502** may comprise silicon nitride, silicon dioxide (SiO₂), silicon oxy-nitride (e.g., SiON), or the like. An upper inter-level dielectric (ILD) layer **202d** is subsequently formed over the upper insulating layer **1502**. The upper insulating layer **1502** has a first side facing the substrate **102** and a second side that faces the upper ILD layer **202d**.

(64) As shown in cross-sectional view **1600** of FIG. **16**, the upper ILD layer **202d** is selectively patterned to form an opening **1602**. The opening **1602** is defined by sidewalls of the upper ILD layer **202d**, the upper insulating layer **1502**, and the hard mask layer **1304**. The opening **1602** vertically extends from a top of the upper ILD layer **202d** to the top electrode **118** of the RRAM device **111**.

(65) As shown in cross-sectional view **1700** of FIG. **17**, an upper interconnect structure **120** is formed within the opening **1602** in the upper ILD layer **202d**. In some embodiments, the upper interconnect structure **120** may comprise an upper interconnect wire or via. In some embodiments, the upper interconnect structure **120** may be formed by filling the opening **1602** with a conductive material (e.g., copper and/or aluminum) and subsequently performing a planarization process (e.g., a chemical mechanical planarization process) to remove an excess of the conductive material from over the upper ILD layer **202d**.

(66) FIG. **18** illustrates a flow diagram of some embodiments of a method **1800** of forming an integrated chip having an RRAM device.

(67) While method **1800** is illustrated and described below as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

(68) At **1802**, an access transistor is formed within a substrate. FIG. **7** illustrates a cross-sectional view **700** of some embodiments corresponding to act **1802**.

(69) At **1804**, one or more lower interconnect layers are formed within one or more stacked ILD layers over the substrate. FIG. **8** illustrates a cross-sectional view **800** of some embodiments corresponding to act **1804**.

(70) At **1806**, a lower insulating layer is formed over the one or more lower interconnect layers and the one or more stacked ILD layers. The lower insulating layer comprises an opening that extends through the lower insulating layer to the one or more lower interconnect layers. FIG. **9** illustrates a cross-sectional view **900** of some embodiments corresponding to act **1806**.

(71) At **1808**, a bottom electrode structure is formed over the lower insulating layer and the one or more lower interconnect layers. FIG. **10** illustrates a cross-sectional view **1000** of some embodiments corresponding to act **1808**. In some embodiments, the bottom electrode structure may be formed according to acts **1810-1812**. At **1810**, a bottom electrode diffusion barrier film is formed over the plurality of lower interconnect layers and the lower insulating layer. At **1812**, a bottom electrode metal film is subsequently formed over the bottom electrode diffusion barrier film.

(72) At **1814**, a diffusion barrier film is formed over the bottom electrode structure. FIG. **11** illustrates a cross-sectional view **1100** of some embodiments corresponding to act **1814**.

(73) At **1816**, a dielectric data storage film is formed over the diffusion barrier film. FIG. **12** illustrates a cross-sectional view **1200** of some embodiments corresponding to act **1816**.

(74) At **1818**, a capping film is formed over the dielectric data storage film in some embodiments. FIG. **12** illustrates a cross-sectional view **1200** of some embodiments corresponding to act **1818**.

(75) At **1820**, a top electrode structure is formed over the capping film. FIG. **12** illustrates a cross-sectional view **1200** of some embodiments corresponding to act **1820**.

(76) At **1822**, the top electrode structure is selectively patterned to define top electrode. FIG. **13** illustrates a cross-sectional view **1300** of some embodiments corresponding to act **1822**.

(77) At **1824**, dielectric spacers are formed on opposing sides of the top electrode. FIG. **14** illustrates a cross-sectional view **1400** of some embodiments corresponding to act **1824**.

(78) At **1826**, the dielectric data storage film, the diffusion barrier film, and the bottom electrode structure are selectively patterned to define a dielectric data storage layer, a diffusion barrier layer, and a bottom electrode. FIG. **14** illustrates a cross-sectional view **1400** of some embodiments corresponding to act **1826**.

(79) At **1828**, an upper inter-level dielectric (ILD) layer is deposited over the top electrode. FIG. **15** illustrates a cross-sectional view **1500** of some embodiments corresponding to act **1828**.

(80) At **1830**, an upper interconnect structure is formed onto the top electrode. FIGS. **16-17** illustrate cross-sectional views **1600-1700** of some embodiments corresponding to act **1830**.

(81) Although the disclosed figures and description are described in relation to resistive random access memory (RRAM) devices, it will be appreciated that the disclosed diffusion barrier layer is not limited to such memory devices. Rather, in some alternative embodiments, the disclosed diffusion barrier layer may also be applied to other types of memory devices (e.g., programmable metallization cell memory device, phase change memory device, or the like).

(82) Accordingly, in some embodiments, the present disclosure relates to an integrated chip having an RRAM cell comprising a diffusion barrier layer arranged between a bottom electrode and a dielectric data storage layer. The diffusion barrier layer is configured to prevent the diffusion of metal atoms (e.g., noble metal atoms) from the bottom electrode into the dielectric data storage layer, thereby allowing the dielectric data storage layer to have a substantially uniform thickness that increases reliability of the RRAM device.

(83) In some embodiments, the present disclosure relates to an integrated chip. The integrated chip includes a bottom electrode disposed over one or more interconnect layers; a diffusion barrier layer arranged over the bottom electrode; a data storage layer separated from the bottom electrode by the diffusion barrier layer; and a top electrode arranged over the data storage layer. In some embodiments, the diffusion barrier layer has a bottom surface directly contacting the bottom electrode and a top surface directly contacting the data storage layer. In some embodiments, the diffusion barrier layer includes tantalum nitride, titanium nitride, titanium carbon nitride, or tungsten nitride. In some embodiments, the diffusion barrier layer includes a noble metal oxide or a noble metal nitride. In some embodiments, the bottom electrode includes a bottom electrode diffusion barrier and a bottom electrode metal, and the bottom electrode metal is arranged directly between the diffusion barrier layer and the bottom electrode diffusion barrier. In some embodiments, the bottom electrode metal includes a noble metal. In some embodiments, the top electrode and the bottom electrode diffusion barrier include a same material. In some embodiments, the integrated chip further includes an insulating layer arranged over one or more stacked inter-level dielectric (ILD) layers overlying the substrate, the bottom electrode diffusion barrier having a curved lower sidewall surrounded by the insulating layer and a substantially flat upper sidewall over the insulating layer. In some embodiments, the diffusion barrier layer has a thickness in a range of between approximately 5 Angstroms and approximately 30 Angstroms. In some embodiments, the diffusion barrier layer has a maximum width that is larger than a width of a top surface of the data storage layer.

(84) In other embodiments, the present disclosure relates to an integrated chip. The integrated chip includes one or more interconnect layers arranged within one or more stacked inter-level dielectric (ILD) layers over a substrate; a bottom electrode diffusion barrier laterally surrounded by an insulating layer disposed over the one or more stacked ILD layers; a bottom electrode metal having a bottom surface contacting the bottom electrode diffusion barrier, the bottom electrode metal including a noble metal; a diffusion barrier layer contacting a top surface of the bottom electrode metal; a dielectric data storage layer contacting the diffusion barrier layer; and a top electrode

separated from the bottom electrode metal by the dielectric data storage layer and the diffusion barrier layer. In some embodiments, the top electrode and the bottom electrode diffusion barrier comprise a first material that is different from a second material of the bottom electrode metal. In some embodiments, the top electrode and the bottom electrode diffusion barrier include titanium nitride, tantalum nitride, tungsten, or tungsten nitride. In some embodiments, the diffusion barrier layer has an inner region surrounded by an outer region, the inner region having a recessed upper surface arranged laterally between and vertically below upper surfaces of the outer region. In some embodiments, the diffusion barrier layer has a substantially equal thickness between outermost sidewalls. In some embodiments, the integrated chip further includes dielectric spacers arranged along sidewalls of the top electrode, the dielectric spacers having a horizontally extending segment that protrudes outward from a sidewall of the dielectric spacers.

(85) In yet other embodiments, the present disclosure relates to a method of forming an integrated chip. The method includes forming a bottom electrode structure over one or more interconnect layers, the one or more interconnect layers disposed within one or more stacked inter-level dielectric (ILD) layers over a substrate; forming a diffusion barrier film over the bottom electrode structure; forming a dielectric data storage film onto the diffusion barrier film; forming a top electrode structure over the dielectric data storage film; and patterning the top electrode structure, the dielectric data storage film, the diffusion barrier film, and the bottom electrode structure to define a memory device. In some embodiments, the method further includes forming the diffusion barrier film to a thickness that prevents formation of hillocks including noble metal atoms from the bottom electrode structure from extending into the dielectric data storage film. In some embodiments, forming the bottom electrode structure includes forming a bottom electrode diffusion barrier film over the one or more interconnect layers; and forming a bottom electrode metal film in contact with a top surface of the bottom electrode diffusion barrier film. In some embodiments, the method further includes forming the diffusion barrier film in contact an upper surface of the bottom electrode metal film; and forming the dielectric data storage film in contact with an upper surface of the diffusion barrier film.

(86) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An integrated chip, comprising: a bottom electrode structure disposed over a lower interconnect within a lower inter-level dielectric (ILD) layer over a substrate, wherein the bottom electrode structure comprises an upper surface having a noble metal; a diffusion barrier layer over the bottom electrode structure; a data storage structure over the diffusion barrier layer; a top electrode structure over the data storage structure; and wherein the diffusion barrier layer is configured to mitigate a diffusion of noble metal atoms from the bottom electrode structure to the data storage structure.
2. The integrated chip of claim 1, further comprising: a dielectric material arranged along one or more sidewalls and over an upper surface of the top electrode structure; and an upper ILD layer arranged along sidewalls and over an upper surface of the dielectric material.
3. The integrated chip of claim 2, wherein the dielectric material physically contacts the one or more sidewalls and the upper surface of the top electrode structure.
4. The integrated chip of claim 2, further comprising: an upper interconnect structure extending

- through the upper ILD layer and the dielectric material to contact the top electrode structure.
5. The integrated chip of claim 4, wherein the dielectric material laterally contacts the upper interconnect structure.
 6. The integrated chip of claim 2, wherein the dielectric material has a bottommost surface that is over the diffusion barrier layer.
 7. The integrated chip of claim 2, wherein the dielectric material physically contacts a sidewall and an upper surface of the data storage structure.
 8. The integrated chip of claim 1, further comprising: a bottom electrode diffusion barrier arranged vertically between the bottom electrode structure and the lower interconnect, wherein the diffusion barrier layer has a greater width than a bottom surface of the bottom electrode diffusion barrier.
 9. An integrated chip, comprising: a first electrode structure disposed over a substrate, wherein the first electrode structure comprises a surface having a noble metal; a diffusion barrier on the surface of the first electrode structure; a switching structure disposed on the diffusion barrier and being configured to store data, wherein the diffusion barrier is configured to mitigate a diffusion of noble metal atoms from the first electrode structure into the switching structure; and a second electrode structure on the switching structure.
 10. The integrated chip of claim 9, wherein the diffusion barrier has an inner upper surface that is arranged laterally between and vertically below an outer upper surface of the diffusion barrier, the outer upper surface wrapping around the inner upper surface in a plan-view of the diffusion barrier.
 11. The integrated chip of claim 10, further comprising: an insulating structure laterally surrounding the first electrode structure, wherein the first electrode structure continuously extends from over the insulating structure to a bottom of the insulating structure.
 12. The integrated chip of claim 9, wherein the diffusion barrier has a larger thickness measured through a center of the diffusion barrier than along an outermost sidewall of the diffusion barrier.
 13. The integrated chip of claim 9, further comprising: a sidewall spacer contacting the diffusion barrier and being arranged along outermost sidewalls of the switching structure and the second electrode structure.
 14. The integrated chip of claim 9, wherein the diffusion barrier comprises ruthenium or iridium.
 15. The integrated chip of claim 9, wherein the diffusion barrier has a surface physically contacting the noble metal along an interface.
 16. An integrated chip, comprising: a first electrode structure comprising a first electrode diffusion barrier and a noble metal on the first electrode diffusion barrier; a diffusion barrier layer on the first electrode structure, the diffusion barrier layer being separated from the first electrode diffusion barrier by the noble metal; a data storage structure on the diffusion barrier layer; and a second electrode structure on the data storage structure.
 17. The integrated chip of claim 16, wherein the diffusion barrier layer has a substantially flat first surface physically contacting the noble metal.
 18. The integrated chip of claim 17, wherein the diffusion barrier layer has a substantially flat second surface physically contacting the data storage structure, the substantially flat second surface opposing the substantially flat first surface.
 19. The integrated chip of claim 16, wherein the data storage structure comprises hafnium oxide, zirconium oxide, aluminum oxide, nickel oxide, tantalum oxide, or titanium oxide.
 20. The integrated chip of claim 16, wherein the diffusion barrier layer completely separates the noble metal from the data storage structure.
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